

[5353]-192

T.E. (I.T.)

THEORY OF COMPUTATION

(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

Q1) a) Consider the following ϵ -NFA

[6]

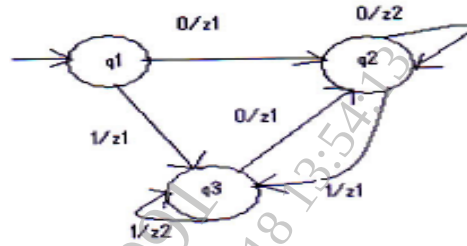
	a	b	c	ϵ
p	{p}	{q}	{r}	ϕ
q	{q}	{r}	ϕ	{p}
r	{r}	ϕ	{p}	{q}

- i) ϵ -closure of each state
 - ii) Give all strings of length 3 or less accepted by the automation.
 - iii) Convert the automation to DFA
- b) Give RE for the following languages over $\Sigma = \{0,1\}$ [4]
- i) The language of all strings containing at least two 0's.
 - ii) The language of all strings containing exactly two 0's.
 - iii) The language of all strings containing every 0 followed by 11.
 - iv) The language of all strings not containing the substring 000.

OR

P.T.O

Q2) a) Construct Moore machine for given Melay machine. [6]



b) By using Pumping Lemma, prove that following language
 $L = \{a^n \mid n \geq 1\}$ is not regular. [4]

Q3) a) Convert given CFG to GNF [6]

$S \rightarrow AB$

$A \rightarrow BS \mid b$

$B \rightarrow SA \mid a$

b) Consider CFG with productions [4]

$S \rightarrow aB \mid bA$

$A \rightarrow a \mid aS \mid bAA$

$B \rightarrow b \mid bS \mid aBB$

For string aaabbabbba

Find leftmost and rightmost derivation.

OR

Q4) a) Convert the following grammar to their equivalent CNF [6]

$S \rightarrow PQP$

$P \rightarrow aP \mid \epsilon$

$Q \rightarrow bQ \mid \epsilon$

b) Convert Left Linear Grammar to equivalent Right Linear Grammar. [4]

$S \rightarrow C0 \mid A0 \mid B1$

$A \rightarrow A1 \mid C0 \mid B1 \mid 0$

$B \rightarrow B1 \mid 1$

$C \rightarrow A0$

- Q5)** a) Write a note on closure properties of CFL's. [4]
- b) Let $L = \{a^m b^n; n < m\}$ Construct [8]
- CFG accepting L
 - PDA accepting L by empty store
 - PDA accepting L by final state
- c) Construct PDA equivalent to the following CFG [6]
- $S \rightarrow 0A10BA$
 $A \rightarrow S010$
 $B \rightarrow 1B1$

OR

- Q6)** a) Give CFG generating the language accepted by following PDA [8]
- $M = (\{q_0, q_1\}, \{a, b\}, \{z_0, X\}, \delta, q_0, z_0, \phi)$, δ is given below
- $\delta(q_0, b, z_0) = \{(q_0, Xz_0)\}$
 $\delta(q_0, b, X) = \{(q_0, XX)\}$
 $\delta(q_0, a, X) = \{(q_1, X)\}$
 $\delta(q_1, b, X) = \{(q_1, \epsilon)\}$
 $\delta(q_1, a, z_0) = \{(q_0, z_0)\}$
- b) Design PM for $L = \{a^n b^n | n \geq 0\}$. [6]
- c) Compare the power of Post machine and Push down Automata. [4]

- Q7)** a) Design a Turing Machine to add two unary numbers. [8]
- b) Explain Universal Turing machine. [4]
- c) Differentiate between FA, PDA and TM. [4]

OR

- Q8)** a) Construct TM for obtaining 2's complement of a binary number. Simulate TM for any input. [8]
- b) Write short note on Halting problem of TM. [8]

- Q9)** a) Explain Post Correspondence Problem with example. [8]
b) Explain recursive language and recursively enumerable language with suitable example. [8]

OR

- Q10)** a) Define decidability of problem. Describe undecidable problems for Context Free Grammar. [8]
b) Write short note on: [8]
i) Turing Reducibility
ii) Multitape TM

